

ICE COMMANDER RIP

Towards the end of August, the first man to have walked the moon breathed his last. While many paid their tributes, The Economist's obituary on Neil Armstrong is one that will stay with you long after you have read it.

<http://dgit.in/OCXCww>

STATE OF THE WEB INDEX

In this Q&A, Sir Tim Berners Lee, explains the objective of the World Wide Web Foundation's latest project – Web Index. It is basically the world's first multi-dimensional measure of the web's growth, utility and impact on people and nations. India is ranked #33 by the way.

<http://dgit.in/RoD5N5>

OUR LIFE, A SIMULATION?

In this one-on-one, NASA scientist Rich Terrile, talks about the possibility that our reality, may in fact be a computer simulation, designed by a programmer from the future.

<http://dgit.in/OEjB6j>

POST-TOUCHSCREEN ERA

This article explores the technologies that will be ubiquitous in the post-touchscreen era of input. It shows video demos of these technologies under test and also has quotes from researchers of projects working on these technologies.

<http://dgit.in/Uc5kil>

iPhone 5 benchmarks

Slower than the Galaxy S3, faster than the Nexus 7s

- By Sebastian Anthony



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Over the weekend, the first iPhone 5 benchmarks have started to appear on Geekbench. The results, if they're real, highlight two very interesting facts: The new core inside the A6 SoC is Apple's own creation, rather than a Cortex-A9 or A15; and despite the new core, the latest dual- and quad-core Android devices are just as fast, or faster.

The iPhone5,2, officially known as model number A1429 (the Verizon model if you're in the US), scores 1601 on Geekbench. According to Geekbench, the iPhone 5 has two ARMv7 cores clocked at 1.02GHz, 1GB of RAM, and the same caches as the A5 (32KB L1 instruction, 32KB L1 data, 1MB L2). The iPhone 4S (A5 SoC) has an average Geekbench score of around 600, so the iPhone 5 is more than twice as fast — which tallies with Apple's performance claims.

To put the iPhone 5/A6 in context: The quad-core Galaxy S3 (Exynos 4 Quad, a Cortex-A9 design) has a Geekbench score of around 1800; the dual-core One X (Snapdragon S4, Krait) scores around 1600; and the quad-core Nexus 7 (Tegra 3, Cortex-A9) scores around 1600. We must also bear in mind that Geekbench makes full use of multiple cores, which gives the quad-core chips an unfair advantage — after all, there are few if any smartphone apps that make good use of four cores. In reality, the A6 will probably perform very similarly to the dual-core Snapdragon S4.



The iPhone 5's benchmarks have two standout features: The A6 is clocked very low (just 1GHz, versus the 1.4-1.5GHz chips in most other smartphones), and its memory performance is out of this world. Compared to the 4S (Cortex-A9), the iPhone 5 has somewhere in the region of 3 to 5 times better memory performance. Even compared to a newer chip, such as the Snapdragon S4, the A6 is still almost twice as fast. Despite the low clock speed, the A6's dual-core CPU still has between two and three times the theoretical performance of the A5, again tallying with Apple's claim of 2x performance. The Snapdragon S4 has slightly better integer and floating point performance, but at a much higher clock speed (1.5GHz). It is the A6's low clock speed and excellent performance that suggests that Apple has designed its own core, rather than using a beefed-up Cortex-A9 or new Cortex-A15. Anandtech is also confirming, through its own research, that the A6 indeed features a custom CPU core.

The benchmarks suggest that the A6 core is similar to Krait — which is wider than Scorpion (the previous Snapdragon S3 core), but without deepening the pipeline. A wider out-of-order architecture allows for an increase in IPC (instructions per clock), but without driving the clock speed (and thus power consumption) up. Remember, Apple has also

increased battery life between the iPhone 4S and iPhone 5, which is partly thanks to Qualcomm's single-chip 28nm radio and the shift to 32nm on the A6, but there's undoubtedly some funky stuff going on inside the SoC too.

Why did Apple design its own core?

Two questions remain: If the A6 SoC does contain a custom-made CPU core, why didn't Apple advertise that fact during the keynote presentation? Furthermore, why did Apple design its own core anyway?

I can only guess at the answer to the first question. It could be that Apple is sworn to secrecy through a licensing deal (maybe Qualcomm designed the core for Apple?), or that it's working on patenting the design. It's possible that yields of the chip (which we presume are being made on Samsung's 32nm process) are low, and that only some iPhone 5s will have the new core inside.

As for why Apple designed its own core, the answer is fairly simple: Apple is a control freak. Its previous SoCs, the A4, A5, and A5X, used conventional Cortex cores — but otherwise they were heavily customized. Designing its own CPU core was just the next logical step. With its own core, and perhaps its own instruction set extensions, Apple can ensure that iOS 6 and the A6 are perfectly synergistic, further increasing performance and lowering battery consumption. There is also the possibility that the Cortex-A15 simply doesn't provide the power efficiency that Apple requires, forcing the company to design its own core that is both efficient and quick.